STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-RESOLVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3185-3197 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3199-3261, the complete author proof (61 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_sdga.json (pb=2700, stmt 3185-3197, proof 3199-3261).

Editorial note. Section 10.2 checked: the region carries a second complete proof environment (the author's 'Second proof', lines 3263-3442, building the resolution as a total complex). Only the FIRST proof (lines 3199-3261) is retained inside the card span, per the gate's one-proof-per-card rule and this lane's precedent; the second proof is an independent alternative and is not counted as a separate card (same chapter + same label).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-resolve' also exists in stacks-dga — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSG: sources/stacks-sdga/pinned-original-source.tex, source lines 3129–3137. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSC: sources/stacks-sdga/pinned-original-source.tex, source lines 2983–2991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSD: sources/stacks-sdga/pinned-original-source.tex, source lines 3022–3029. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0FSF: sources/stacks-sdga/pinned-original-source.tex, source lines 3098–3126. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $\mathcal{M}$ be a differential graded $\mathcal{A}$-module. There exists a homomorphism $\mathcal{P} \to \mathcal{M}$ of differential graded $\mathcal{A}$-modules with the following properties*

(1) *$\mathcal{P} \to \mathcal{M}$ is a quasi-isomorphism, and*
(2) *$\mathcal{P}$ is good.*

First proof. Let $\mathcal{S}_0$ be the sheaf of graded sets (Remark 0FSF) whose degree n part is $\mathrm{Ker}(\mathrm{d}^n_{\mathcal{M}})$. Consider the homomorphism of differential graded modules

$$\mathcal{P}_0 = \mathcal{A}[\mathcal{S}_0] \longrightarrow \mathcal{M}$$

where the left hand side is as in Remark 0FSF and the map sends a local section s of $\mathcal{S}_0$ to the corresponding local section of $\mathcal{M}^{\deg(s)}$ (which is in the kernel of the differential, so our map is a map of differential graded modules indeed). By construction the induced maps on cohomology sheaves $H^n(\mathcal{P}_0) \to H^n(\mathcal{M})$ are surjective. We are going to inductively construct maps

$$\mathcal{P}_0 \to \mathcal{P}_1 \to \mathcal{P}_2 \to \ldots \to \mathcal{M}$$

Observe that of course $H^*(\mathcal{P}_i) \to H^*(\mathcal{M})$ will be surjective for all i. Given $\mathcal{P}_i \to \mathcal{M}$ denote $\mathcal{S}_{i+1}$ the sheaf of graded sets whose degree n part is

$$\mathrm{Ker}(\mathrm{d}^{n+1}_{\mathcal{P}_i}) \times_{\mathcal{M}^{n+1}, \mathrm{d}} \mathcal{M}^n$$

Then we set

$$\mathcal{P}_{i+1} = \mathcal{P}_i \oplus \mathcal{A}[\mathcal{S}_{i+1}]$$

as graded $\mathcal{A}$-module with differential and map to $\mathcal{M}$ defined as follows

(1) for local sections of $\mathcal{P}_i$ use the differential on $\mathcal{P}_i$ and the given map to $\mathcal{M}$,
(2) for a local section $s = (p, m)$ of $\mathcal{S}_{i+1}$ we set $\mathrm{d}(s)$ equal to p viewed as a section of $\mathcal{P}_i$ of degree $\deg(s) + 1$ and we map s to m in $\mathcal{M}$, and
(3) extend the differential uniquely so that the Leibniz rule holds.

This makes sense because $\mathrm{d}(m)$ is the image of p and $\mathrm{d}(p) = 0$. Finally, we set $\mathcal{P} = \mathrm{colim}\, \mathcal{P}_i$ with the induced map to $\mathcal{M}$.

The map $\mathcal{P} \to \mathcal{M}$ is a quasi-isomorphism: we have $H^n(\mathcal{P}) = \mathrm{colim}\, H^n(\mathcal{P}_i)$ and for each i the map $H^n(\mathcal{P}_i) \to H^n(\mathcal{M})$ is surjective with kernel annihilated by the map $H^n(\mathcal{P}_i) \to H^n(\mathcal{P}_{i+1})$ by construction. Each $\mathcal{P}_i$ is good because $\mathcal{P}_0$ is good by Lemma 0FSG and each $\mathcal{P}_{i+1}$ is in the middle of the admissible short exact sequence $0 \to \mathcal{P}_i \to \mathcal{P}_{i+1} \to \mathcal{A}[\mathcal{S}_{i+1}] \to 0$ whose outer terms are good by induction. Hence $\mathcal{P}_{i+1}$ is good by Lemma 0FSC. Finally, we conclude that $\mathcal{P}$ is good by Lemma 0FSD. $\square$